Photosynthesis - Study Notes

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy into chemical energy stored as glucose.

The overall equation is:

$$6\ CO_2 + 6\ H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6\ O_2$$

It occurs mainly in the leaves, inside organelles called chloroplasts. Chloroplasts contain the green pigment chlorophyll, which absorbs mostly red and blue light and reflects green light.

Two Stages of Photosynthesis

1. Light-dependent reactions:
   - Take place in the thylakoid membranes.
   - Water is split (photolysis), releasing oxygen as a by-product.
   - Produce ATP and NADPH.

2. Light-independent reactions (the Calvin cycle):
   - Take place in the stroma of the chloroplast.
   - Use ATP and NADPH to fix carbon dioxide into glucose.
   - This stage does not directly require light.

Factors that affect the rate of photosynthesis include light intensity, carbon dioxide concentration, and temperature.